

PROBLEM STATEMENT

The Vacuum Cleaner Problem is one of the fundamental problems in Artificial Intelligence used to demonstrate the concept of intelligent agents. The environment consists of a set of rooms, each of which may either be **clean** or **dirty**. The vacuum cleaner agent can perceive the state of its current location and perform actions such as **cleaning** the room or **moving** to another room. The objective is to clean all dirty rooms while minimizing the number of actions performed. The problem can be extended to any number of rooms and different room configurations.